F79SP Session 10: Markov chains: Absorption probabilities

1 Absorption probabilities

We know once we are inside a closed class, we cannot leave. What is the probability of being absorbed into a closed class, given that we start from $X_0 = i$ for $i \in S$?

Set up: Let C be a closed class. Define the column vector $\boldsymbol{\alpha}^C = \left(\alpha_1^c, \alpha_2^c, \ldots, \alpha_i^c, \ldots\right)^T$ where:

$$\alpha_i^C = \mathbf{P}(\{X_n\}_{n\geq 0} \text{ eventually absorbed in } C \,|\, X_0 = i)$$

. We can compute $\boldsymbol{\alpha}^C$ as follows.

Theorem 1.1. *1. The vector $\boldsymbol{\alpha}^C$ satisfies the following system of equations:*

$$\begin{aligned} \boldsymbol{\alpha}^C &= P\boldsymbol{\alpha}^C \\ \alpha_i^C &= 1 \qquad \textit{for all } i \in C \\ \alpha_i^C &= 0 \qquad \textit{for all } i \textit{ in any other closed classes.} \end{aligned}$$

2. The vector $\boldsymbol{\alpha}^C$ is the minimum *solution of the above system of equations.*

3. In the case where the set of states in open *classes is* finite *(in particular when $\mathcal{S}$ is* finite*) the vector $\boldsymbol{\alpha}^C$ is the* unique *solution of the above system of equations. Further in this case, for any initial state i, the sum of the absorption probabilities over all closed classes is* 1*.*

Proof. We prove 1 only, using conditioning on first step technique:

Moreover:

- $\alpha_i^C = 0$ for all i in other closed classes, as they can't leave the closed class.
- $\alpha_i^C = 1$ for all $i \in C$ since they were already in C.

□

Example 1.1. *Reconsider the THMC* $\{X_n\}_{n\geq 0}$ *with* $S = \{1, 2, 3\}$ *with transition matrix* $P = (p_{ij})_{i,j\in S}$.

$$P = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Observe that:

- *There is only one closed class* $C = \{1\}$ *and* S *is finite. Hence, it must be that:*

Examples. We consider again some of the earlier examples.

Simple random walk on $\mathcal{S} = \{0, 1, 2, \dots\}$ with 0 an absorbing state. The state 0 forms the single closed class $C = \{0\}$, and all other states form an open class $O = \{1, 2, \dots\}$. Then the probability of absorption in the class C is just the probability of ruin. It is a straightforward exercise to use the above theorem, including the *minimality* condition of part 2, to give a direct proof of the earlier Corollary, i.e.

$$\mathbf{P}_a(R) = \alpha_a^C \equiv \alpha_a = \begin{cases} \left(\frac{q}{p}\right)^a & p > \frac{1}{2} \\ 1 & p \leq \frac{1}{2}. \end{cases}$$

Health insurance system. The health insurance system considered above has a single closed class consisting of the absorbing state D. Since $\mathcal{S}$ is finite it follows from part 3 of the above theorem (or by direct calculation using part 1)that eventual absorption in this state is certain!

Further example. Consider again the example of the previous section with $\mathcal{S} = \{1, 2, 3, 4, 5\}$. Let $\boldsymbol{\alpha}$ be the vector of absorption probabilities in the closed class consisting of the single absorbing state 5. We have $\alpha_1 = \alpha_2 = 0$, $\alpha_5 = 1$, and

$$\begin{aligned} \alpha_3 &= \frac{1}{5}\alpha_3 + \frac{1}{5}\alpha_4 + \frac{1}{5} \\ \alpha_4 &= \frac{1}{3}\alpha_3 + \frac{1}{3}, \end{aligned}$$

giving $\alpha_3 = 4/11$, $\alpha_4 = 5/11$.

It follows directly from part 3 of the above theorem that the vector of absorption probabilities in the closed class $\{1, 2\}$ is $\mathbf{1} - \boldsymbol{\alpha}$.